

Technical Report: Mathematical & Computational Methodology

Project: India Air Quality & Climate Digital Twin

Document Series: Capstone Report Materials — Part 4

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Reference Research: Capstone Research (241).docx (Nature Reviews Earth-System Digital Twin framework; Kawano et al.; CPCB Guidelines)

1. Domain Modeling & Problem Formulation

The overarching goal is to formulate an **Earth-System Digital Twin** for the Indian subcontinent (68.0°E to 97.5°E , 6.0°N to 37.5°N) that harmonizes satellite remote sensing observations, chemical transport reanalysis, and thermodynamics into two tightly coupled learning frameworks:

$$\mathcal{DT} = \langle \mathcal{M}_1, \mathcal{M}_2, \mathcal{S}_t, \mathcal{R}_t, \mathcal{F}_{\text{scenario}} \rangle$$

Where:

- $\mathcal{M}_1$: Multi-pollutant surface $\text{PM}_{2.5}$ and AQI estimation model with HCHO anomaly tracking and fire attribution.
- $\mathcal{M}_2$: Climate digital twin forecaster projecting atmospheric-terrestrial states.
- $\mathcal{S}_t$: Continuous coupled geospatial state tensor.
- $\mathcal{R}_t$: Compound multi-hazard climate risk engine.
- $\mathcal{F}_{\text{scenario}}$: Interactive counterfactual simulation operator.

2. Model 1 Methodology: Surface AQI & Atmospheric Chemistry

2.1 Spatial Discretization & Multi-Sensor Fusion

The continuous land area of India is mapped onto a regular grid with spatial resolution $\Delta\theta = 0.25^\circ \approx 28\text{ km}$:

$$\mathcal{G} = \{ (\lambda_i, \phi_j) \mid \lambda_i = 68.0^\circ + i \cdot 0.25^\circ, \phi_j = 6.0^\circ + j \cdot 0.25^\circ \}$$

For each cell (i, j) and daily epoch $t \in [1, 730]$ (2022–2023), multi-spectral features are fused:

1. Satellite Remote Sensing:

- Sentinel-5P TROPOMI: Vertical column densities (VCD) for NO_2 , CO , SO_2 , O_3 , HCHO with strict quality flag screening ($\text{qa_value} > 0.5$, cloud fraction < 0.3).
- NASA MODIS: MAIAC AOD_{550} , Land Surface Temperature (LST_{day} , $\text{LST}_{\text{night}}$), and vegetation index (NDVI).

2. Atmospheric Chemical Reanalysis (MERRA-2):

- $\text{PM}_{2.5}$ dry mass at 35% relative humidity ($\mu\text{g}/\text{m}^3$), total AOD, black carbon, dust, and sulfate mass mixing ratios.

3. Planetary Boundary Layer & Meteorology (ECMWF ERA5):

- 2m temperature (T_{2m}), relative humidity (RH), 10m wind speed vector (U_{10} , V_{10}), surface pressure (SP), total precipitation (TP), boundary layer height (BLH).
- Ventilation Index (VI): Quantifying pollutant dispersal capacity:

$$VI = BLH \times \sqrt{U_{10}^2 + V_{10}^2} \quad [\text{m}^2/\text{s}]$$

4. Fire Radiative Power & Proximity Metrics (NASA VIIRS FIRMS):

- Aggregated fire counts and sum of Fire Radiative Power (FRP, MW) evaluated across circular geodesic buffers $B_r \in \{25\text{ km}, 50\text{ km}, 100\text{ km}, 200\text{ km}\}$ with temporal transport lags $\tau \in \{0, 1, 2\text{ days}\}$:

$$FRP_{\{r, \tau\}}(i, j, t) = \sum_{k \in K} FRP_k \cdot \mathbb{I}(\text{dist}(p_k, (i, j)) \leq r)$$

2.2 Piecewise Linear CPCB National AQI Engine

The ground truth calibration is governed by the Central Pollution Control Board (CPCB) National Air Quality Index standard:

$$I_p = \frac{I_{\text{high}} - I_{\text{low}}}{B_{\text{high}} - B_{\text{low}}} \left(C_p - B_{\text{low}} \right) + I_{\text{low}}$$

Where the piecewise breakpoints for 24-hr $\text{PM}_{2.5}$ ($\mu\text{g}/\text{m}^3$) are:

- Good (0–50): $0 \leq C \leq 30$
- Satisfactory (51–100): $31 \leq C \leq 60$
- Moderate (101–200): $61 \leq C \leq 90$
- Poor (201–300): $91 \leq C \leq 120$
- Very Poor (301–400): $121 \leq C \leq 250$
- Severe (401–500): $251 \leq C \leq 500+$

2.3 Formaldehyde (HCHO) Hotspot Identification & Clustering

Formaldehyde acts as an atmospheric marker for non-methane volatile organic compound (NMVOC) emissions:

1. Seasonal Anomaly Z-Score:

$$Z_{\{i,j,t\}} = \frac{\Omega_{\text{HCHO}}(i, j, t) - \mu_{\text{season}}(i, j)}{\sigma_{\text{season}}(i, j)}$$

- Thresholding & DBSCAN Clustering: Cells with $Z \geq 2.0$ are clustered using Density-Based Spatial Clustering of Applications with Noise ($\epsilon = 1.0$, $\text{MinPts} = 3$).

3. Multi-Source Attribution: Co-location with **NO₂** indicates industrial VOCs; co-location with active fires indicates agricultural burning; and co-location with high **NDVI** denotes biogenic isoprene emission.

3. Model 2 Methodology: Climate Digital Twin & Scenario Engine

3.1 Coupled State Space & Multi-Step Forecasting

The coupled state $\mathbf{S}_t \in \mathbb{R}^{H \times W \times C}$ encapsulates the thermodynamic and chemical state of the subcontinent. Multi-horizon sequence models predict:

$$\mathbf{S}_{t+1:t+K} = \mathcal{G}_{\phi}(\mathbf{S}_{t-L:t})$$

Where:

- Lookback window $L = 14$ days.
- Lead horizons $K \in \{7, 14, 30\}$ days.
- Implemented via a deep Sequence-to-Sequence **LSTM** (PyTorch) and multi-output **Gradient Boosted Decision Trees** (XGBoost).
- **Skill Score Benchmark:** Evaluated against the standard persistence forecast:

$$\text{Skill Score} = 1 - \frac{\text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{persistence}}}$$

Achieved **SS = 0.875** at 7 days and **0.894** at 30 days.

3.2 Compound Multi-Hazard Risk Formulation

Traditional single-variable indices fail to capture joint climate crises. The digital twin computes a composite risk surface:

$$\mathcal{R}_{\text{composite}}(i, j, t) = w_1 \mathcal{R}_{\text{heat}} + w_2 \mathcal{R}_{\text{drought}} + w_3 \mathcal{R}_{\text{AQ}}$$

Where weights are calibrated to $w_1 = 0.35$, $w_2 = 0.30$, $w_3 = 0.35$ and:

1. **Heat Stress ($\mathcal{R}_{\text{heat}}$):** Wet-bulb globe temperature (**WBT**) thresholding combined with daytime MODIS **LST** anomalies exceeding the 90th climatological percentile:

$$\mathcal{R}_{\text{heat}} = \text{clip}\left(\frac{T_{2m} - 28.0}{14.0} + 0.3 \cdot \frac{\text{LST} - \mu_{\text{LST}}}{\sigma_{\text{LST}}}, 0, 1\right)$$

2. **Drought Vulnerability ($\mathcal{R}_{\text{drought}}$):** Rolling 30-day cumulative precipitation deficit relative to normal monsoon levels compounded by vegetative stress ($\Delta \text{NDVI} < -0.15$).
3. **Air Quality Exceedance ($\mathcal{R}_{\text{AQ}}$):** Daily probability of surface concentrations exceeding the National Ambient Air Quality Standard ($\text{PM}_{2.5} > 60 \mu\text{g}/\text{m}^3$) or reaching the Severe category ($\text{PM}_{2.5} > 250 \mu\text{g}/\text{m}^3$).

3.3 Counterfactual Scenario Simulation Engine

To explore policy and climate interventions, the digital twin operates as a fast response surface $\hat{\mathbf{f}}_{\theta}$:

$$\mathbf{S}' = \mathbf{S} + \mathcal{T}(\Delta T_{2m}, \Delta \text{TP}, \Delta \text{Fire}, \Delta \text{Emissions})$$

- Evaluates thermodynamic stagnation during heatwaves.
- Quantifies aerosol scavenging during enhanced or deficit rainfall.
- Simulates public health benefits of targeted stubble burning abatement programs across Punjab and Haryana.